Dynamics of structures

Laboratory 2: Multiple Degrees of Freedom System

Professor:
Francisco Hernandez
Assistance:
Gastón Parra
Students:
Bernardo Caprile
Pedro Valenzuela
Felipe Vicencio
Lukas Wolff

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1 Introduction

Understanding the dynamic behavior of real-world structures requires the analysis of systems with multiple degrees of freedom (MDOF). Unlike single degree of freedom (SDOF) models, which offer only a simplified representation, MDOF systems provide a more realistic framework for simulating buildings, bridges, and mechanical structures subjected to dynamic loads such as wind, traffic, and earthquakes. These systems are characterized by multiple natural frequencies and associated mode shapes, each contributing uniquely to the overall structural response.

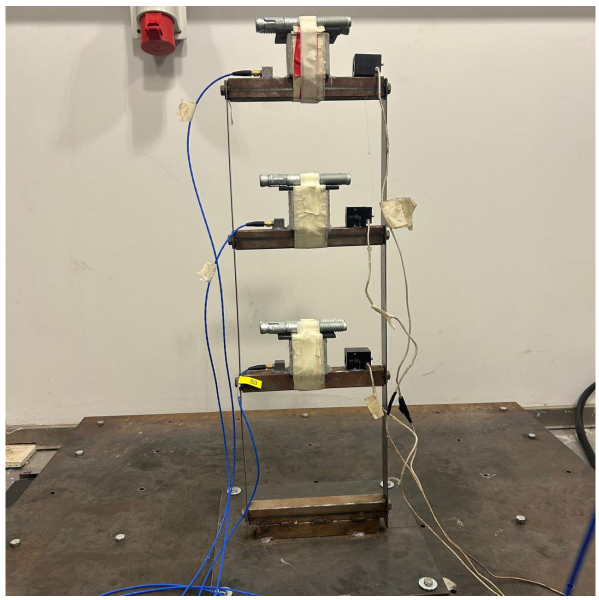


Figure 1: Representation of an MDOF structural model

This laboratory experiment focuses on the dynamic analysis of an MDOF system subjected to various excitation conditions. The primary objectives are:

- To identify the natural frequencies and corresponding mode shapes of a 3-degree-of-freedom structure through analytical and experimental methods.
- To evaluate the system's response under free vibration (pullback tests) and harmonic excitation, including the influence of damping and resonance.
- To analyze the seismic response of the system under two real earthquake records (Kobe and Concepción), using both analytical simulations and experimental data.

This comprehensive study builds upon the theoretical foundations of SDOF systems and extends them to more complex configurations. It provides critical insights into modal analysis, dynamic amplification, and the effect of damping essential concepts for the safe and effective design of real engineering structures subjected to dynamic loads.

2 Theoretical Background

2.1 Natural Frequency

The natural frequency of a system is the rate at which the body oscillates when it is disturbed without any damping or external forces acting on it. This parameter is crucial in understanding the dynamic behavior of structures, as it determines how the system will respond to external excitations. The natural frequency can be calculated using the following formula:

$$\omega_n = \sqrt{\frac{k}{m}} \tag{1}$$

Where:

- ω_n = natural frequency (rad/s)
- k = stiffness of the system (N/m)
- m = mass of the system (kg)

One of the important aspects of the natural frequency is that it is inversely proportional to the mass of the system. This means that as the mass increases, the natural frequency decreases, and vice versa. This relationship is crucial in designing structures to ensure they can withstand dynamic loads without resonating at their natural frequency.

2.2 Damping Ratio

The damping ratio is a parameter that has the effect of reducing the oscillations of a system over time. In physical systems, damping is caused by energy dissipation mechanisms such as friction, air resistance or dampers. The damping ratio is defined as the ratio of the actual damping to the critical damping, and it can be calculated using the following formula:

$$\beta = \frac{c}{2\sqrt{mk}} \tag{2}$$

Where:

- β = damping ratio (dimensionless)
- c = damping coefficient (N.s/m)
- k = stiffness of the system (N/m)
- m = mass of the system (kg)

2.3 Logarithmic Decrement

The logarithmic decrement is a measure of the rate at which the amplitude of oscillations decreases over time in a damped system. It is defined as the natural logarithm of the ratio of two successive amplitudes, and it can be calculated using the following formula:

$$\delta = \frac{1}{n}\ln\left(\frac{x_0}{x_n}\right) \tag{3}$$

Where:

- δ = logarithmic decrement (dimensionless)
- x_0 = amplitude of the first oscillation (m)
- x_n = amplitude of the n-th oscillation (m)
- n = number of oscillations (dimensionless)

With this parameter, we can determine the damping ratio of the system using the following formula:

$$\beta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}} \tag{4}$$

Where:

- δ = logarithmic decrement (dimensionless)
- β = damping ratio (dimensionless)

2.4 Harmonic base excitation

The harmonic base excitation is a phenomenon that occurs when a system is subjected to a periodic force at its base. This type of excitation is key in understanding the dynamic behavior of structures, due to every structure suffers from this type of excitation.

2.5 Transmissibility

Transmissibility is a measure of how much the input motion is transmitted to the output of a system. It is defined as the ratio of the output response to the input response, and it can be calculated using the following formula:

$$TR = \frac{\ddot{X}}{\ddot{X}_g} = \sqrt{\frac{1 + (2\beta(\bar{\omega}/\omega_n)^2}{(1 - \frac{\bar{\omega}^2}{\omega_n^2})^2 + (2\beta\frac{\bar{\omega}}{\omega_n})^2}} \tag{5}$$

Where:

- TR = transmissibility (dimensionless)
- $\ddot{X}_g$ = output acceleration (m/s^2) (acceleration of the base)
- $\ddot{X}$ = input acceleration (m/s^2) (acceleration of the mass)
- $\bar{\omega}$ = frequency of the input motion (rad/s)
- ω_n = natural frequency of the system (rad/s)
- β = damping ratio (dimensionless)

Due to the transmissibility, if the excitation frequency is much smaller than the natural frequency of the system ($\bar{\omega} < \omega_n$), the transmissibility is close to 1, which means that the input motion is transmitted to the output with little amplification. On the other hand, if the excitation frequency is larger than the natural frequency of the system ($\bar{\omega} > \omega_n$), the transmissibility is close to 0, which means that the input motion is transmitted to the output with null amplification. Finally, if the excitation frequency is equal to the natural frequency of the system ($\bar{\omega} = \omega_n$), the transmissibility is infinite, which means that the input motion is transmitted to the output with a high amplification. This phenomenon is known as resonance.

2.6 Resonance

Resonance is a phenomenon that occurs when the frequency of an external force matches the natural frequency of a system:

$$\omega_n = \bar{\omega}$$

For an undamped system the factor of amplification is infinite, which means that the system will oscillate indefinitely with increasing amplitude. In real systems, damping reduces the maximum amplitude of oscillations, but resonance can still lead to significant increases in amplitude. This means, if the resonance is not controlled, it can cause structural damage or failure.

2.7 Mass Matrix and Stiffness Matrix

In multi-degree-of-freedom (MDOF) systems, the dynamic behavior is governed by the equation of motion:

$$\mathbf{M}\ddot{\mathbf{X}}(t)+\mathbf{C}\dot{\mathbf{X}}(t)+\mathbf{K}\mathbf{X}(t)=\mathbf{F}(t) \tag{6}$$

Where:

- $\mathbf{M}$ = mass matrix (kg)
- $\mathbf{C}$ = damping matrix (kg/s)
- $\mathbf{K}$ = stiffness matrix (N/m)
- $\mathbf{X}(t)$ = displacement vector
- $\mathbf{F}(t)$ = external force vector

The mass matrix $\mathbf{M}$ is a diagonal matrix in most lumped mass models, where each diagonal term represents the mass associated with a degree of freedom. The stiffness matrix $\mathbf{K}$ defines the relationship between nodal displacements and restoring forces in the system. Its entries depend on the stiffness properties of structural members.

Where:

- ω = natural angular frequency (rad/s)
- ϕ = mode shape vector (eigenvector)

Each mode shape is associated with a unique natural frequency, and the set of mode shapes forms a complete, linearly independent basis for the system's motion.

2.8 Eigenvalues and Eigenvectors

Solving the eigenvalue problem yields:

- **Eigenvalues** $\lambda_i=\omega_i^2$, which represent the squared natural frequencies of the system.
- **Eigenvectors** ϕ_i, which represent the mode shapes corresponding to each frequency.

The eigenvalue problem for free vibration (undamped) systems is:

$$\mathbf{K}\phi_i=\lambda_i\mathbf{M}\phi_i \tag{7}$$

By taking the square root of the eigenvalues, we obtain the natural frequencies $\omega_i=\sqrt{\lambda_i}$. These values provide critical information about the dynamic response of the structure and are used to identify potential resonance conditions.

2.9 Mode Shapes

The mode shapes (or eigenvectors) of a system represent the relative displacements of each degree of freedom during free vibration at a specific natural frequency. They are determined by solving the eigenvalue problem of the undamped system:

$$\left(\mathbf{K}-\omega^2\mathbf{M}\right)\phi=\mathbf{0} \tag{8}$$

3 Structure Description

The structure analyzed in this laboratory corresponds to a three-degree-of-freedom (3DOF) system subjected to free vibration. It consists of three vertically aligned masses connected by steel beams that provide bending stiffness. Each mass is allowed to move in the vertical direction, and the structure mimics a simplified building model with lumped masses and flexible connections. The dynamic response was captured using accelerometers placed on each mass.

Masses:
The total mass of the system is concentrated in three rigid blocks, each formed by stacked steel components and instrumentation. Additionally, the self-weight of the connecting beams is included in the equivalent mass at each level.

The structure follows the assumption that masses are not tributary: each degree of freedom is associated with a single lumped mass, which receives half the weight of the beams above and half of those below. Therefore, the equivalent lumped masses for each degree of freedom are:

- $m_1 = 0.555 + 0.26 + 4 \times \frac{0.075}{6} = 0.829$ kg
- $m_2 = 0.553 + 0.26 + 8 \times \frac{0.075}{6} = 0.993$ kg
- $m_3 = 0.691 + 0.26 + 8 \times \frac{0.075}{6} = 1.131$ kg

These values include half of the mass of each adjacent beam, distributed based on the position of each mass in the structure.

The mass matrix **M** (in Tonf·s^2/m) is:

$$\mathbf{M} = \begin{bmatrix} 8.45 \times 10^{-5} & 0 & 0 \\ 0 & 1.01 \times 10^{-4} & 0 \\ 0 & 0 & 1.15 \times 10^{-4} \end{bmatrix}$$

Stiffness:
The stiffness of each level is governed by the flexural stiffness of the steel beams, modeled as fixed-fixed beams. Each column consists of two metallic rulers (steel strips) working in parallel, which increases the total stiffness of each connection.

The stiffness of a single beam is calculated as:

$$k_{\text{single}} = \frac{48EI}{L^3}$$

Since each connection is composed of two identical rulers acting in parallel, the total stiffness per level becomes:

$$k_i = 2 \times k_{\text{single}} = \frac{96EI}{L^3}$$

Where:

- $E = 200\,\text{GPa}$ is the Young's modulus of steel,
- $I = \frac{1}{12}bh^3$ is the second moment of area,
- $L = 0.6/3 = 0.2\,\text{m}$ is the span of each beam.

With this, the stiffness per level is:

$$k_i = \frac{96EI}{L^3} \approx 110{,}371.4\,\text{Tonf/m}$$

The global stiffness matrix $\mathbf{K}$ (in Tonf/m) is:

$$\mathbf{K} = \begin{bmatrix} 110371.4 & -110371.4 & 0 \\ -110371.4 & 220742.8 & -110371.4 \\ 0 & -110371.4 & 220742.8 \end{bmatrix}$$

Natural Frequencies:
The natural frequencies of the system were computed using the eigenvalue problem $\mathbf{K}\phi = \omega^2\mathbf{M}\phi$. The eigenvalues were obtained, and then the natural frequencies were calculated as:

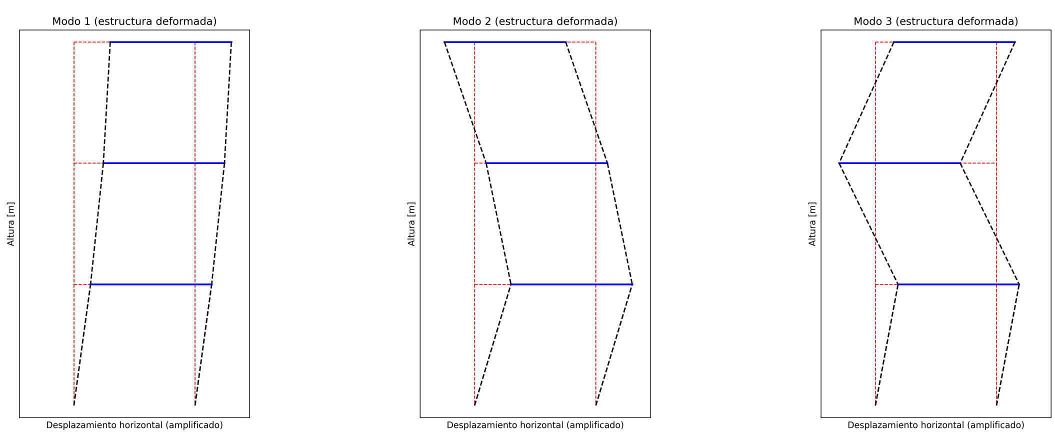
$$\omega_n = \sqrt{\lambda_n}, \quad f_n = \frac{\omega_n}{2\pi}$$

The computed natural frequencies are:

$$\omega_1 = 14.06\,\text{rad/s}, \quad \omega_2 = 37.96\,\text{rad/s}, \quad \omega_3 = 55.57\,\text{rad/s}$$
$$f_1 = 2.24\,\text{Hz}, \quad f_2 = 6.04\,\text{Hz}, \quad f_3 = 8.84\,\text{Hz}$$

Theoretical Mode Shapes

The theoretical vibration modes of the 3DOF system are shown below. These represent the deformation patterns associated with each natural frequency computed from the analytical model.



Mode 1 Mode 2 Mode 3

Figure 2: Theoretical mode shapes of the 3DOF system.

4 Instrumentation Details

4.1 Endevco 44A16 Accelerometer

The Endevco model 44A16 accelerometer is a triaxial IEPE (Integrated Electronics Piezo-Electric) sensor, designed for general-purpose vibration measurement applications, such as the dynamic analysis of the SDOF structure carried out in this laboratory.

This model has a sensitivity of 100 mV/g and a measurement range of ± 50 g, which allows it to record significant accelerations without saturating the signal. In addition, it offers a useful frequency response range between 0.6 Hz and 5000 Hz depending on the axis, enabling it to capture the dynamic responses of various systems.

4.2 QuantumX CX22B-W Data Acquisition System

The QuantumX CX22B-W data acquisition system by HBM is a high-performance standalone data logger designed to capture, analyze, and store mechanical, electrical, or thermal variables without the need to be connected to an external computer. It is particularly useful for capturing data in structural analysis, as it can process data in real time and operate with multiple channels.

This device is equipped with a quad-core Intel Atom E3845 processor running at 1.9 GHz, 4 GB of RAM, and 240 GB of internal storage, in addition to a slot for a CFast card of up to 512 GB, which allows for extended data recording periods. It supports a maximum total sampling rate of 5 MS/s, making it capable of capturing high-frequency dynamic phenomena. Through its various input options, the system can integrate multiple types of sensors such as GPS units, cameras, and force sensors, among others.

For data analysis and visualization, the software *Catman Easy* version 5.1.3 was used. This program enables the configuration of measurements and real-time data visualization, and allows automatic analysis such as the Fast Fourier Transform (FFT). In this case, acceleration responses were recorded and the FFT results were displayed every 5 seconds.

4.3 TDG-250 KG Shaking Table

The TDG-250 KG shaking table, manufactured by Teknik Destek Grubu, is a uniaxial seismic motion simulator designed for conducting structural dynamic tests in laboratory settings. Its maximum dynamic load capacity is 250 kg when applying an acceleration of 1 g, while for lighter loads of up to 100 kg, it can reach accelerations of up to ± 2 g. The table has a maximum stroke of ± 200 mm, a peak velocity of 800 mm/s, and an operational frequency of 15 Hz, making it suitable for a variety of dynamic input signals. Its displacement resolution is 0.0025 mm, providing high precision in motion control.

The system is driven by a servomotor and controlled via a closed-loop PID controller. It operates with a three-phase power supply of 3.3 kW at 380–480 V AC and 50 or 60 Hz.

The shaking table is operated using the *Easy Test Shake Table* software, which allows the user to import and reproduce real earthquake records, such as those from Concepción, Chile or Kobe, Japan, as well as generate custom signals. The software supports various types of dynamic inputs, and during testing, it provides real-time visualization of acceleration, velocity, and displacement profiles, along with Fast Fourier Transform (FFT) results and response spectra.

5 Signal Processing

Signal processing is carried out to prepare raw signals obtained from the sensors prior to analysis. This process includes two main techniques: detrending and digital filtering. Detrending involves removing constant components or linear trends from the signal, which may result from initial displacements, systematic errors, or very low-frequency noise. This technique allows the signal to be centered around zero, a necessary condition for computing parameters such as velocity and displacement.

Next, digital filtering is used to eliminate unwanted components from the signal, such as high-frequency noise or low-frequency elements that are not part of the structural response being analyzed.

Both techniques are implemented using the *Catman Easy* software version 5.1.3, in conjunction with the QuantumX CX22B-W data acquisition system by HBM. This setup enables the recording and real-time visualization of the measured signals, as well as the application of digital processing during or after the test. Additionally, the software allows for the automatic visualization of the Fast Fourier Transform (FFT) at defined intervals, enabling the identification of dominant frequencies and the analysis of the dynamic behavior of the structure.

6 Laboratory Methodology

The objective of this laboratory was to analyze the dynamic behavior of a multiple-degree-of-freedom (MDOF) structural system. To achieve this, a shaking table, acceleration sensors, and a data acquisition system were used.

First, a pull-back test was performed, which consisted of applying an initial lateral displacement to the structure and then releasing it to observe its free response. This test allowed the determination of the system's natural periods and damping ratios through the analysis of the acceleration signal using the logarithmic decrement method. The Fast Fourier Transform (FFT) of the acceleration response was computed in real time every 5 seconds using the Catman Easy software (version 5.1.3).

Subsequently, a harmonic loading test was conducted, in which the shaking table was programmed to generate sinusoidal motions with specific frequencies and amplitudes. This test enabled the observation of the resonance phenomenon by analyzing how the response amplitude varies with the excitation frequency. The harmonic signals were selected to excite the first three vibration modes of the structure.

Finally, a seismic test was carried out using real earthquake records. The records of the Kobe and Concepción (27F) earthquakes were scaled and reproduced on the shaking table. The aim was to observe the system's response to complex excitations and to assess its structural behavior under realistic seismic events.

Data collection was performed using an Endevco 44A16 accelerometer connected to the QuantumX CX22B-W data acquisition system. The signals were sampled at a frequency of 200 Hz, which allowed for precise recording of the system's dynamic response over time. The *Catman Easy* software version 5.1.3 was used to record, visualize, and process the signals in real time, including the computation of the FFT.

Post-processing of the data enabled the identification of the system's dynamic parameters and characterization of its behavior under different types of excitation.

7 Results

7.1 Pull-back Test

Determination of the damping period, natural period and damping ratio

To determine the natural frequencies and damping characteristics of the system, the logarithmic decrement method was applied to the acceleration signals obtained from the pull-back test (Pullback 2). This method allows estimation of the damping ratio ζ and the damped natural period T_d by evaluating the exponential decay of the vibration envelope.

Since the system under analysis is a multiple-degree-of-freedom (MDOF) structure, three acceleration signals were acquired, corresponding to each degree of freedom (DOF 1, DOF 2, and DOF 3). The signals were detrended and filtered to remove noise and baseline shifts. The processed time-domain responses are shown in Figure 3.

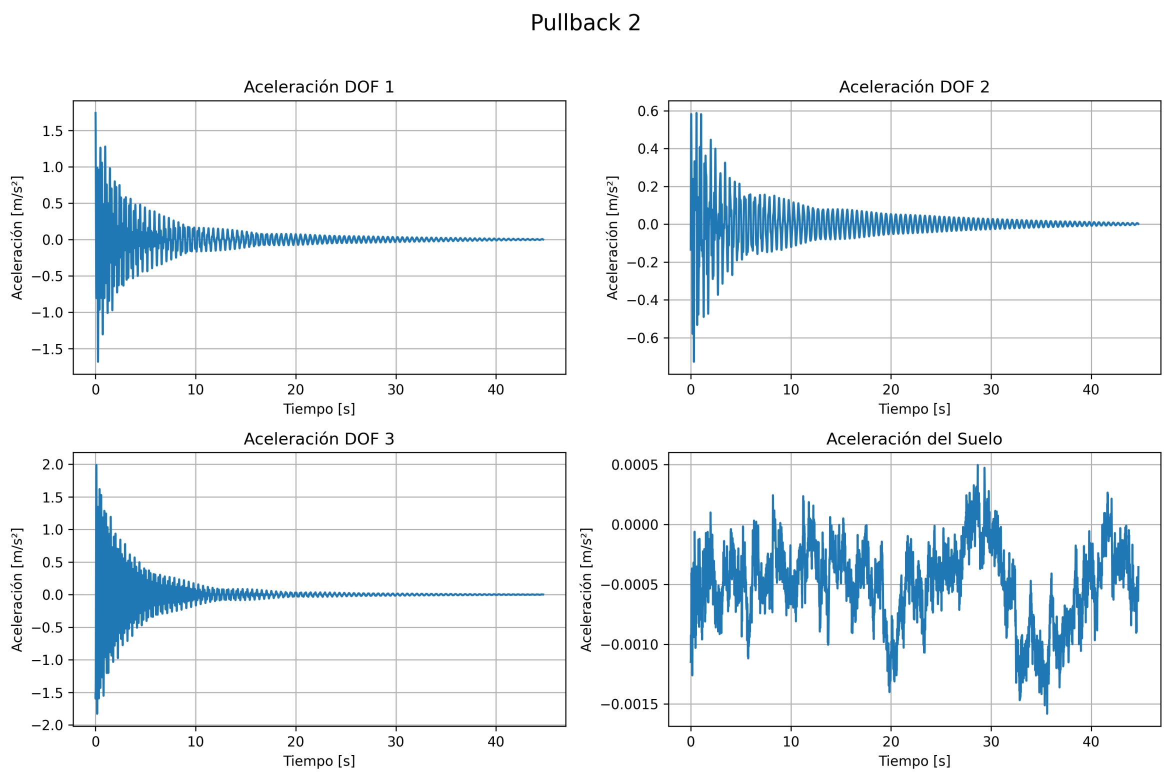


Figure 3: Acceleration response of each DOF and ground during Pullback 2.

To identify the dominant frequencies excited during the free vibration, the Fast Fourier Transform (FFT) was computed for each DOF. The results are presented in Figure 4, where three distinct peaks can be observed for each signal, corresponding to the natural frequencies of the MDOF system.

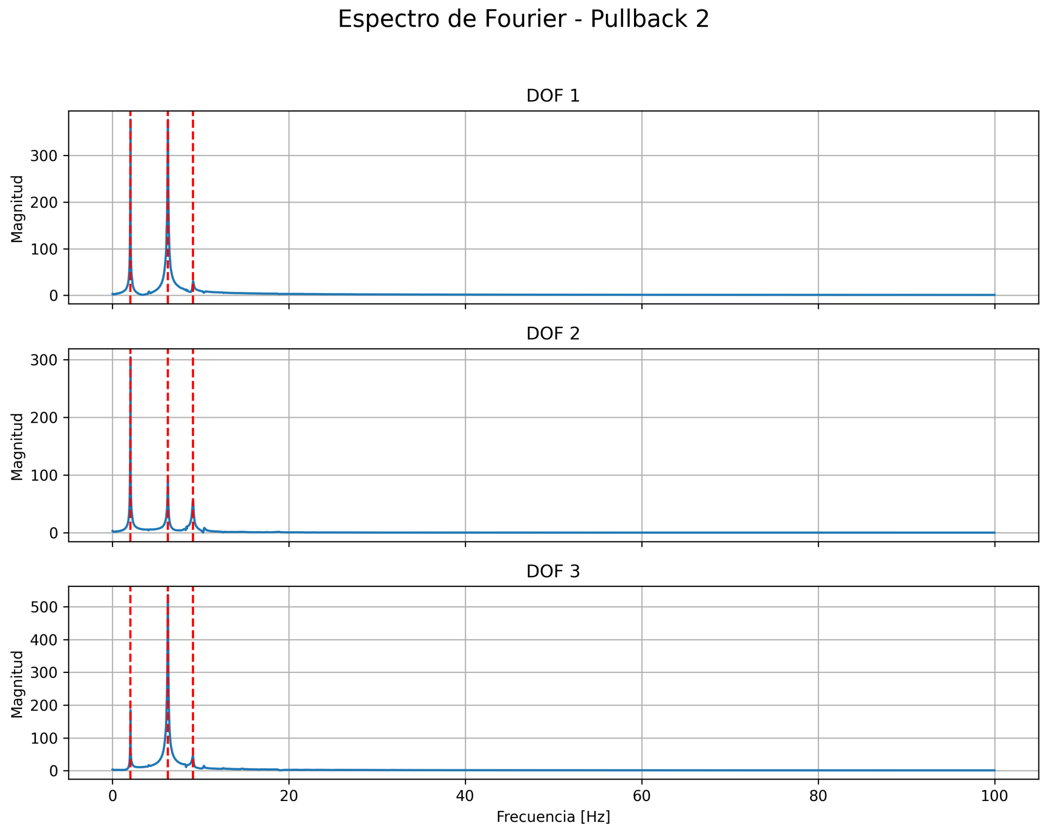


Figure 4: Fourier spectrum of Pullback 2 for each DOF. Vertical dashed lines indicate the identified natural frequencies.

The dominant frequencies identified from the FFT for each degree of freedom are:

- **DOF 1:** 2.06 Hz, 6.31 Hz, 9.15 Hz
- **DOF 2:** 2.06 Hz, 6.28 Hz, 9.15 Hz
- **DOF 3:** 2.06 Hz, 6.31 Hz, 9.12 Hz

Using the decay of successive peaks in the time-domain response, the logarithmic decrement was applied to estimate the damping ratio ζ. The damped natural period T_d was calculated from the inverse of each dominant frequency. Table 1 summarizes the estimated values:

Mode	f_n (Hz)	T_d (s)	ζ
Mode 1	2.06	0.485	0.0114
Mode 2	6.30	0.159	0.0114
Mode 3	9.14	0.109	0.0114

Table 1: Natural frequencies, damped periods, and damping ratio estimated from Pullback 2.

As shown in the table, the structure exhibits three well-defined natural frequencies, with low damping ratios across all modes. These results confirm that the system behaves as a lightly damped MDOF structure. The consistent exponential decay and frequency peaks support the validity of the linear model used for modal analysis.

Comparison between Theoretical and Experimental Frequencies

The natural frequencies obtained experimentally from the pull-back test (Pullback 2) were compared with the theoretical values computed from the analytical model of the MDOF system.

Table 2 summarizes the theoretical and experimental frequencies for the first three modes, along with the absolute difference and the relative error percentage.

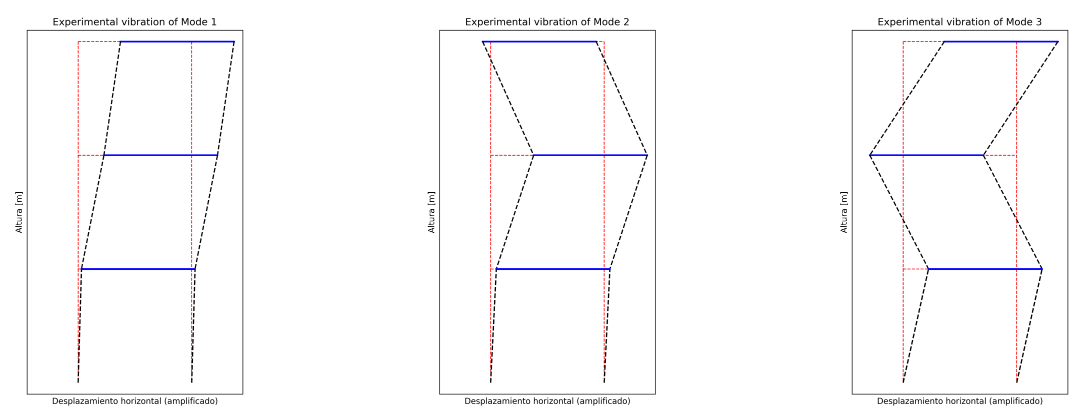
Mode	Theoretical f_n [Hz]	Experimental f_n [Hz]	Error [%]
Mode 1	2.24	2.06	8.04%
Mode 2	6.04	6.30	4.30%
Mode 3	8.84	9.14	3.39%

Table 2: Comparison between theoretical and experimental natural frequencies.

As seen in Table 2, the relative error between theoretical and experimental frequencies is below 9% in all modes, which is considered acceptable given the assumptions made in the analytical model (e.g., ideal boundary conditions, perfectly linear behavior, simplified stiffness estimation). The experimental values tend to be slightly lower in the first mode and slightly higher in the higher modes, which may be due to non-idealities such as damping, assembly imperfections, or mass distribution effects.

Test with harmonic vibrations

For this section, a comparison was made between the theoretical and experimental modes of vibrations of the strcuture when resonance occurs.



(a) VIbration Mode 1 (b) Vibration Mode 2 (c) Vibration Mode 3

Figure 5: Experimental vibration modes of the structure during resonance in harmonic excitation.

Table 3: Natural experimental frequencies of the structure

Mode	f_n [Hz]
1	2.06
2	6.3
3	9.14

Table 4: Normalized experimental mode shapes of the structure

Mode	Floor 1	Floor 2	Floor 3
1	0.10	0.76	1.24
2	0.17	1.26	-0.23
3	0.75	-0.98	1.21

Table 5: Normalized theoretical modes of the structure

Mode	Floor 1	Floor 2	Floor 3
1	0.57	1.00	1.24
2	0.36	0.11	-0.30
3	0.07	-0.12	0.06

As Table 3, 4 and 5 shows, some differences are observed in both the frequencies and the mode shapes. These discrepancies can be attributed to factors such as model simplifications, experimental noise, or limitations in capturing the full dynamic behavior of the structure. Overall, the results highlight the effectiveness of the experimental approach while also indicating areas where further refinement of the model or the testing procedures may be beneficial.

Despite these differences, the vibration modes plotted were in accordance with the theoretical modes, confirming the expected behavior of the structure under harmonic excitation.